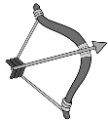


Problem #3: Bow or Onion? (15 marks)

Words often have multiple meanings. For example, the English word **crane** can either mean 'a bird' or 'a machine used for lifting heavy objects'. Similarly, the Russian word **лук** (pronounced as **luk**) has two meanings: **onion** and **bow** (a weapon). In the first paragraph below the word **luk** means **onion** whereas in the second paragraph it means **bow**.



shalot i **luk** (**onion**) ochen tesno svjazannye ovoschi. oba imejut lukovichnye rastenija. v vkusa, shalot bolee ostrij i, sledovatelno, ne edjat v kolichestvakh.



bak gorazdo bolee prodvinityj, oruzie, chem **luk** (**bow**). vmesto togo, chtoby streljali iz lukov ona ispolzuetsja, chtoby streljat snarjadami. eti snarjady ispolzujutsja v sovremennoj vojne, chtoby ubit vragov.

For each of the following paragraphs containing the word **luk**, identify whether it means **onion** or **bow**? Explain how you arrived at your conclusions. For ease of reading, the Cyrillic letters used for writing Russian have been mapped to similar sounding Latin letters.

1. pistoljet javljaetsja gorazdo bolee sovremennij oruzie, chem luk. vmesto togo, chtoby strely iz lukov on ispolzuetsja dlja metanija puljami. on ispolzuetsja v sovremennoj okhoty ubit zivotnykh.
2. chesnok i luk tesno svjazannye ovoschi . oni oba imejut lukovichnye rastenie. v vkusa, chesnok bolee ostrij i, sledovatelno, ne edjat v bolshikh kolichestvakh.
3. okhotnikov ispolzovat luk iz dereva dlja okhoty na razlichnykh zivotnykh. oni chasto imel navyki v strelbe iz luka i praktika ego s samogo rannego vozrasta. oni ispolzujut luk, metanija strely i ubit nichego ne podozrevajuschikh zertv.
4. luk luchshe kultiviruetsja v plodorodnykh pochvakh, kotorye khorosho drenirovannye. v otlichie ot mnogikh drugikh ovoschey, luk trebujut vysokogo urovnja pitatelnykh veschestv v pochve. rastenie trebuje 11-12 chasov solnechnogo sveta.
5. luk soderezit paru izognutykh uprugikh konechnostej iz dereva. obe konechnosti soedineny stroki. okhotniki specialistam v strelbe iz luka tjanet stroku nazad, chtoby metanija.
6. luk ispolzuetsja kak ovosch i javljaetsja naibolee shiroko kultiviruemykh i edjat vidov. eto naibolee chasto raz v dva goda ili mnogoletnee rastenie. ego listja golubovato - zelenogo cveta.
7. luk i chesnok obychno ispolzujutsja v kachestve prigotovlenija pischi ingredienta dlja povysenija vkus indijskikh karri. on ispolzuetsja v kachestve zagustitelja dlja karri. on takze edjat kak salat vmeste s drugimi ovoschami.
8. luki i strely byli shiroko ispolzuetsja v primitivnoj vojny. v sovremennoj vojne, oni byli zameneny orudij i raket. khotja strelba iz luka bolshe ne trebuetsja v sovremennoj vojne ne do sikh por praktika kak vid sporta.
9. luk i strely predstavljajet soboj sistemu snarjad oruzie. strelba iz luka eto iskusstvo, praktika ili iskusstvo ispolzovanii luka dlja metanija strel. eto ochen primitivnoj sistema ispolzuetsja okhotnikov i peschernogo cheloveka.
10. luk marinovannyj v uksuse edjat v kachestve zakuski. ego listja inogda ispolzuet dlja podgotovki indijskikh karri. luk takze mozno upotrebljat v pischu syrye otlichie chesnok, kotoryj takze javljaetsja ovosch prinadlezaschikh k odnomu vidu.

Choosing the correct meaning of a word in a given context (sentence or paragraph) is an important task which must be solved for developing language processing and understanding software such as automatic translation systems. This task is known as *Word sense disambiguation*. The cues that a computer uses to solve this task is probably similar to what you will be using to solve this problem!